

RESEARCH PROPOSAL

Knowledge Discovery and Data Mining (CIS 4523/5523)

Spring 2026

(0) STUDENT INFORMATION

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(1) TITLE

Detecting Factual Hallucinations in Retrieval-Augmented Generation Systems: A Comparative Study of Faithfulness Detection Methods

(2) OBJECTIVE AND SIGNIFICANCE

The objective of this project is to develop and compare methods for detecting factual hallucinations in Retrieval-Augmented Generation (RAG) systems. Specifically, we aim to determine whether generated answers are faithful to (supported by) their source documents.

This work addresses a critical barrier to deploying RAG systems in production environments. While RAG aims to ground language model outputs in retrieved documents, models frequently generate content that contradicts or goes beyond the provided sources—a phenomenon known as hallucination. This is particularly problematic in high-stakes domains such as healthcare, legal services, and financial advising, where factual accuracy is paramount and misinformation can have serious consequences.

The significance of this project lies in: 1) Providing a systematic comparison of hallucination detection approaches accessible to practitioners; 2) Identifying which methods work best for different types of factual errors; 3) Contributing benchmarked results that can inform deployment decisions for real-world RAG applications; 4) Demonstrating whether off-the-shelf models can effectively detect hallucinations without extensive fine-tuning.

(3) BACKGROUND

Retrieval-Augmented Generation (RAG) combines information retrieval with language generation to produce answers grounded in external knowledge sources. Despite this architecture, large language models (LLMs) still exhibit hallucination—generating plausible-sounding but factually incorrect content.

Recent research categorizes hallucinations into intrinsic (contradicting the source) and extrinsic (adding unsupported information). Our focus is on faithfulness detection: given a question, source document, and generated answer, classify whether the answer is supported by the source. Prior work has explored Natural Language Inference (NLI) models, fine-tuned classifiers, and LLM-as-judge methods. However, few studies provide systematic comparisons of these methods' accuracy, speed, and generalization across datasets. This project fills that gap by benchmarking multiple approaches on standardized datasets.

(4) PROPOSED APPROACH

Data Sources:

We will use three established hallucination detection datasets: 1) HaluEval (primary dataset): 35,000 examples available via Hugging Face, covering multiple domains with hallucination labels; 2) FEVER (Fact Extraction and Verification): 145,000 fact-checking examples for cross-dataset evaluation; 3) BEGIN Benchmark: A challenging dataset with fine-grained annotations for testing model robustness.

Preprocessing Requirements:

Minimal preprocessing is needed as datasets are pre-labeled. Text normalization (lowercasing, removing special characters), format conversion to standardized input: "[QUESTION] {q} [DOCUMENT] {d} [ANSWER] {a}", and train/validation/test splits (70/15/15 for HaluEval).

Methodology:

We will implement and compare three detection methods: Method 1 (NLI Baseline - Zero-Shot): Use pre-trained NLI models (facebook/bart-large-mnli, microsoft/deberta-v3-large-mnli) requiring no training. Method 2 (Fine-Tuned Classifier): Fine-tune microsoft/deberta-v3-small on HaluEval for binary classification to learn hallucination-specific patterns. Method 3 (LLM-as-Judge - Optional): Prompt Claude or GPT-4 to evaluate faithfulness as an upper-bound benchmark.

Evaluation Metrics:

Accuracy, Precision, Recall, F1-score, per-class performance, inference speed, and cross-dataset generalization (train on HaluEval, test on FEVER).

Analysis Plan:

Confusion matrix analysis, error categorization (numerical facts, dates, named entities, reasoning-based), speed-accuracy tradeoff analysis, and statistical significance testing between methods.

Timeline:

Week 1-2 (Mar 30 - Apr 12): Download and preprocess HaluEval dataset, implement zero-shot NLI baseline, begin fine-tuning DeBERTa classifier. Week 3-4 (Apr 13 - Apr 25): Complete fine-tuning, evaluate on FEVER for cross-dataset generalization, conduct error analysis, deliver progress presentations (Apr 18 & 25). Week 5 (Apr 26 - May 2): Optional LLM-judge experiments, finalize all evaluations, write and submit final report.

(5) REFERENCES

- [1] Li, J., Cheng, X., Zhao, W. X., Nie, J. Y., & Wen, J. R. (2023). HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models. *In Proceedings of EMNLP 2023*. <https://arxiv.org/abs/2305.11747>
- [2] Manakul, P., Liusie, A., & Gales, M. J. (2023). SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. *In Proceedings of EMNLP 2023*. <https://arxiv.org/abs/2303.08896>
- [3] Zhang, Y., et al. (2023). Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models. *arXiv preprint arXiv:2309.01219*. <https://arxiv.org/abs/2309.01219>
- [4] Ji, Z., et al. (2023). Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12), 1-38. <https://arxiv.org/abs/2202.03629>